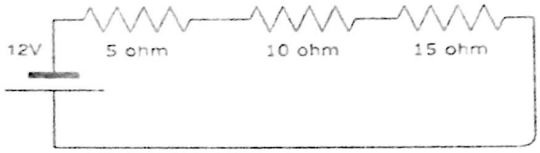
Lab Quiz Examination
Dept. of EEE
Course Name: Physics Sessional, Course Code: PHY 1112
Time: 15 min

Part-I

Marks: 15 (1×15 = 15) **ID NO: 25010617**

1. KVL is applied in _____
   a) Mesh analysis b) Nodal analysis c) Both mesh and nodal d) Neither mesh nor nodal
2. Every_____ is a _____ but every _____ is not a _____
   a) Mesh, loop, loop, mesh b) Loop, mesh, mesh, loop c) Loop, mesh, loop, mesh
   d) Mesh, loop, mesh, loop
3. . Calculate the voltage across the 10 ohm resistor.



   a) 12V b) 4V c) 10V d) 0V
4. KVL deals with the conservation of?
   a) Mass b) Momentum c) Charge d) Energy
5. The sum of the voltages over any closed loop is equal to _____
   a) 0V b) Infinity c) 1V d) 2V
6. 5. The sum of the voltages over any closed loop is equal to _____
   a) 0V b) Infinity c) 1V d) 2V
7. An electric kettle with 50-Ω heat element is powered by a 230-V wall outlet. The current flowing through kettle is:

   (a) 0.217 A (b) 4.6 A (c) 10.86 A (d) 11500 A
8. A junction or a point where two (or more) network elements intersect is called as:
   (a) Node (b) Branch (c) Loop (d) Mesh

9. Which of the following formulas defines KCL, where i1, i2 represent incoming currents and i3, i4 represent outgoing currents from a circuit node?
   (a) i1+i2=i3+i4 (b) i1+i3=i2+i4 (c) i1-i2=i3-i4 (d) i4-i1=i2+i3

10. Kirchhoff law calculates which of the given electric parameters?
    (a) Voltage (b) Resistance (c) Current (d) Both a and c
11. The resistance of an electric circuit is calculated using which of the following laws?
    (a) Kirchhoff's voltage law (b) Kirchhoff's current law (c) Ohm's law (d) Both a and b

CamScanner

12. When 5, 10, and 15 resistors are connected in series with a 12V voltage source, what is the value voltage across resistor 5?
(a) 12 V (b) 21 V (c) 15 V (d) 2 V

13 In a parallel-configured circuit, the current value is:
(a) High (b) Low (c) Zero (d) Infinity

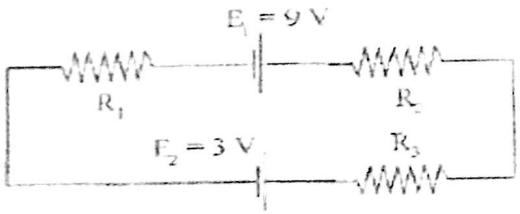
14. In the presence of a magnetic field, are Kirchhoff's laws applicable?
(a) True (b) False (c) None

15. An unknown resistor has 5 mA current flowing through it when 10 volts are applied to it. For the same resistor, the application of 20 volts will result in the current flow of:
(a) 2 mA (b) 5 mA (c) 10 mA (d) 50 mA

Part-II

Marks: 15 **Time: 20 min**

1. State and explain Ohm's Law with its limitations. 1+2 = 3
2. Can you apply KCL and KVL to AC circuits? Justify this statement. 3
3. Draw the circuit diagram of KVL and KCL. 1+1 = 2
4. Is Ohm's Law universal? 2
5. If R1 = 2Ω, R2 = 4Ω, R3 = 6Ω, determine the electric current flows in the circuit below: 5



CamScanner

Lab Quiz Examination

Dept. of EEE

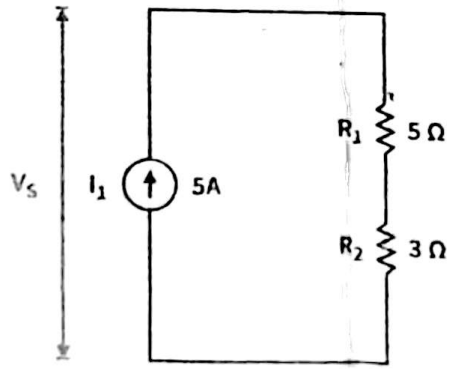
Course Name: Physics Sessional, Course Code: PHY 1112

Time: 20 min

Written Part-II

Marks: 15 **Time: 20 min**

1. What are the other names for KCL ? 1
2. Can you apply KCL to both DC and AC circuits? Justify this statement. 2
3. Is KVL applicable to both series and parallels circuits, Explain it? 2
4. Can KCL be applied to circuits with only one node? 1
5. Consider the following circuit with resistors R1 = 5 ohms and R2 = 3 ohms and a current source I1 = 5A. Determine the voltage drop across the current source using KVL. 4



CamScanner

Department of Electrical & Electronic Engineering (EEE)

Class Test-01

Course Code: PHY 1111, Course Title: Physics

Marks: 10 **Time: 20 Minutes**

1. Define the terms (i) electric dipole (ii) electric flux 2
2. An electric dipole consists of two opposite charges of magnitude 2.0×10^{-6} coulomb separated by a distance, $l = 1.0$ cm. Calculate the maximum torque that the field exert on the dipole having the electric field of 2.0×10^{5} N/C. 3
3. Write down the Coulomb's law. How can you deduce the Coulomb's law from the Gauss' Law. 1+4 = 5

CamScanner

Department of Electrical & Electronic Engineering (EEE)

Class Test-02

Course Code: PHY 1111, Course Title: Physics

Marks: 10 **Time: 20 Minutes**

1. Derive an expression for the capacitance of a parallel-plate capacitor. 4
2. The parallel plates of an air-filled capacitor are 0.5 cm apart. What must be the area of each plate if the capacitance is to be 0.5 F ? 3
3. Define dielectric materials. What happen when a dielectric materials are present into a capacitor? 3